5. DEVELOP A CUSTOMIZED PING COMMAND TO TEST THE SERVER CONNECTIVITY

AIM:

~~To~~ develop a customised ping utility in python that checks the connectivity between a client and a server by sending ICMP Echo request packets and displaying the response time, packet statistics and server reachability status.

PROCEDURE:

1) Install python and VS code.

2) Create ping.py

3) Import subprocess and platform module.

4) Accept the Hostname and IP address from a user.

5) Detect OS using platform module.

6) PING COMMAND:

- ping -n 4 <host> (WINDOWS)
- ping -c 4 <host> (LINUX)

7) Use the subprocess.run() function to execute the ping command and capture the output.

8) Display the ping results.

9) Display "Server status: reached" if successful else "server status: unreachable".

RESULT:

The customized ping program was executed successfully. It checked the connectivity to the specified server and displayed reachable/unreachable, along with ping details.

ping.py - C:/Users/Admin/Downloads/ping.py (3.12.10)

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```
import subprocess
import platform

print("="*50)
print(" Customized Ping Utility")
print("="*50)

host = input("Enter Website or IP Address: ")

system = platform.system()

if system == "Windows":
 command = ["ping", "-n", "4", host]
else:
 command = ["ping", "-c", "4", host]

try:
 result = subprocess.run(command, capture_output=True, text=True)

 print("\nPing Results\n")
 print(result.stdout)
 if result.returncode == 0:
  print("Server Status : Reachable")
 else:
  print("Server Status : Unreachable")

except Exception as e:
 print("Error:", e)
```

IDLE Shell 3.12.10

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```
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr  8 2025, 12:21:36) [MSC v.1943 64 bit
 (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
---------- RESTART: C:/Users/Admin/Downloads/ping.py ----------
==================================================
 Customized Ping Utility
==================================================
Enter Website or IP Address: google.com

Ping Results


Pinging google.com [142.251.222.174] with 32 bytes of data:
Reply from 142.251.222.174: bytes=32 time=5ms TTL=118
Reply from 142.251.222.174: bytes=32 time=5ms TTL=118
Reply from 142.251.222.174: bytes=32 time=4ms TTL=118
Reply from 142.251.222.174: bytes=32 time=4ms TTL=118

Ping statistics for 142.251.222.174:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 5ms, Average = 4ms

Server Status : Reachable
>>>
```